

Phenotype Conflation and Partner-Conditioned Evaluation in Zero-Shot Protein-Protein Interaction Benchmarks

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Zero-shot single-chain protein language models (PLMs) such as ESM-1v, ESM-2, and ESM-Cambrian (ESMC) are frequently benchmarked on deep mutational scanning (DMS) binding assays, achieving reported rank correlations $\rho \approx 0.35\text{--}0.48$. These correlations are often interpreted as evidence that single-sequence masked language modeling captures intermolecular binding physics and quaternary contact energetics. In this work, we demonstrate that benchmark performance can be strongly confounded by target-intrinsic expression and folding stability. In high-throughput selection assays, mutations that destabilize monomer folding impair cell-surface presentation or cellular abundance, indirectly eliminating partner binding and inflating zero-shot scores through monomer-survival prediction. Through an exhaustive provenance audit of deep mutational scanning benchmarks, we show that widely pooled datasets conflate direct display-binding measurements with expression-uncoupled assays, cross-study merges, and cellular fitness proxies. In strictly matched display assays where monomer surface expression and complex binding are measured on the identical mutant library (SARS-CoV-2 RBD and HLA-A*02:01), sequence-only PLM binding correlations at quaternary interfaces attenuate substantially once monomer expression is controlled ($\rho_{\text{partial}} \approx 0.00\text{--}0.14$). In contrast, 3D structure-conditioned models (Protein-MPNN) retain robust interface binding signal ($\rho_{\text{partial}} = +0.202$). Finally, we demonstrate that evaluating single-chain PLMs on single-partner binding readouts cannot distinguish target viability from partner-specific binding, and we formalize a four-stage protocol for partner-conditioned specificity benchmarks.

1 Introduction

Transformer-based protein language models (PLMs) trained on evolutionary sequence databases via masked language modeling—such as ESM-2 (Lin et al. 2023) and ESM-Cambrian (ESMC) (Hayes et al. 2025)—have emerged as standard tools for zero-shot variant effect prediction and protein engineering (Rives et al. 2021; Meier et al. 2021; Hie et al. 2022). In standard evaluation protocols, single-chain masked log-odds ratios $\Delta \log p(x)$ are computed across deep mutational scanning (DMS) datasets compiled in public benchmark suites such as ProteinGym (Notin et al. 2023) and SKEMPI 2.0 (Jankauskaitė et al. 2019).

On these benchmarks, zero-shot PLMs and evolutionary models (Lin et al. 2023; Rao et al. 2021) are reported to achieve moderate-to-high rank correlations on protein-protein interaction (PPI) and complex

binding affinity assays ($\rho \approx 0.35\text{--}0.48$) (Notin et al. 2023). These correlations have fostered the assumption that Transformer architectures trained exclusively on single-chain sequences implicitly capture the biophysics of intermolecular quaternary contacts and binding energetics (Notin et al. 2023; Sledzieski et al. 2021).

In this work, we show that this assumption is vulnerable to a systematic experimental confound: high-throughput binding selection platforms (e.g., yeast surface display, mammalian display, and FACS-based enrichment) conflate **monomer folding stability and cell-surface presentation** with **intermolecular binding affinity** (Figure 1). When a mutation destabilizes the target fold, the protein misfolds and fails surface display. Consequently, no fluorescent binding partner can bind to the absent monomer, producing an apparent loss of binding signal even at positions that make zero contact with the partner.

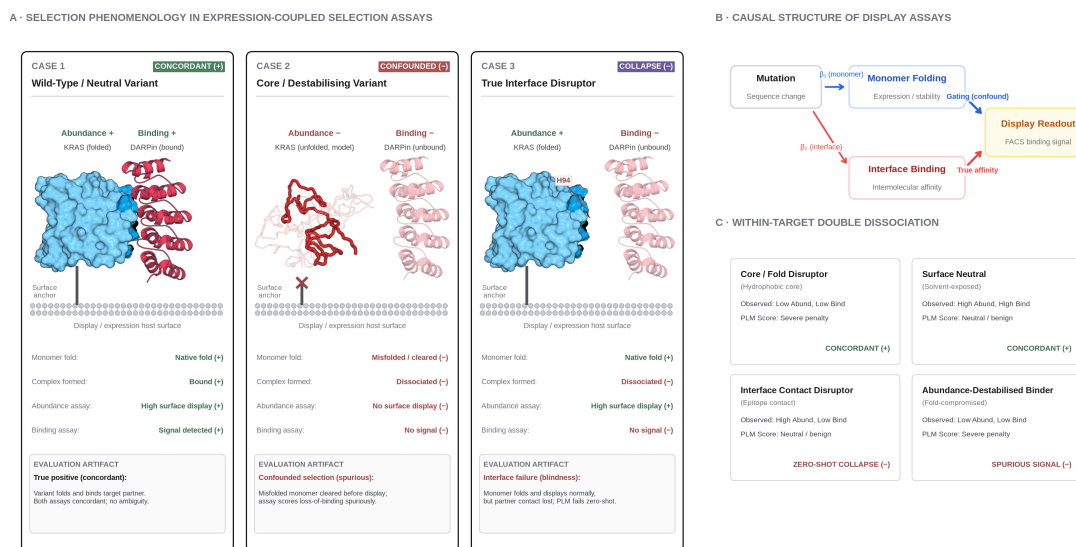


Figure 1: The monomer-folding confound in zero-shot PPI benchmarks. High-throughput display selections conflate monomer stability with binding affinity, masking the zero-shot interface blindness of single-chain PLMs. **(A)** Selection phenomenology in expression-coupled assays, illustrated with the KRAS G-domain–DARPin K55 complex (PDB 6H46). Target monomer rendered as a molecular surface with its contact epitope shaded; partner as a ribbon cartoon. *Case 1*: the folded monomer is presented on the host cell surface and the partner docks onto the epitope (Abundance+, Binding+; concordant true positive). *Case 2*: a core-destabilizing mutation prevents folding, so the monomer is cleared before presentation and no partner can bind (Abundance–, Binding–; confounded selection producing spurious binding correlation). *Case 3*: an interface mutation disrupts partner contact while folding normally (Abundance+, Binding–; zero-shot PLMs score the mutation as benign, failing interface prediction). **(B)** Causal structure of expression-coupled selections, showing that cell-surface presentation gates fluorescent readout. **(C)** Within-target double-dissociation quadrant matrix separating core folding from quaternary interface energetics.

Furthermore, an audit of widely cited benchmark datasets reveals that reported multi-system evaluations frequently pool heterogeneous experimental modalities: strictly matched surface-display assays, cell-growth-based abundance reporters, uncoordinated cross-study merges, and cellular fitness proxies (Table 1). We formalize an auditable provenance tiering for PPI benchmarks, evaluate sequence-only and structure-conditioned models under expression-adjusted controls, and define the experimental requirements for genuine partner-conditioned specificity benchmarking.

Table 1: Audited provenance and experimental classification of primary PPI benchmark systems. Only systems with verified matched expression and binding assays on identical constructs and libraries substantiate strict expression-adjusted binding evaluation.

System ID	Target Complex	Target / Partner Chains	Abundance / Expression Assay	Binding Assay	Provenance Status	Rationale
SARS-CoV-2 RBD	6M0J	E / A (human ACE2)	SPIKE_- SARS2_- Starr_- 2020_- expression	SPIKE_- SARS2_- Starr_- 2020_- binding	Included (Strict Matched)	Simultaneous dual-channel FACS measuring yeast surface expression and ACE2 binding on identical library (Starr et al. 2020).
HLA-A*02:01	5OPI	A / B (β_2M), C (TAPBPR)	Q53Z42_- HUMAN_- McShan_- 2019_- expression	Q53Z42_- HUMAN_- McShan_- 2019_- binding-TAPBPR	Included (Strict Matched)	Dual-staining yeast display measuring surface expression and TAPBPR chaperone binding on identical library (McShan et al. 2019).

System ID	Target Complex	Target / Partner Chains	Abundance / Expression Assay	Binding Assay	Provenance Status	Rationale
KRAS G-domain	6H46	A / B (DARPin K55)	RASK_- HUMAN_- Weng_- 2022_- abundance	RASK_- HUMAN_- Weng_- 2022_- binding- DARPin_K55	Conditional (PCA Growth)	Matched DHFR-PCA abundance reporter and DARPin K55 binding in yeast from same study (Weng et al. 2024); growth-based readout.
Protein G B1 (GB1)	1FCC	C / A, B (IgG Fc)	SPG1_- STRSG_Wu_- 2016	SPG1_- STRSG_- Olson_2014	Excluded (Cross-Study Proxy)	Cross-study merge combining two distinct IgG-Fc binding screens (Wu et al. 2016 ; Olson et al. 2014); no monomer abundance assay.
p53 Oligomerization	10LG	A / B, C, D (tetramer)	P53_HU- MAN_Gia- comelli_- 2018_- Null_Nut- lin	P53_HU- MAN_Gia- comelli_- 2018_WT_- Nutlin	Excluded (Fitness Proxy)	Cancer cell viability under Nutlin-3 in p53-null vs p53-WT A549 cells (Giacomelli et al. 2018); dominant-negative fitness proxy.

2 The Field-Wide Expression Confound

2.1 Assay Mechanics and Selection Conflation

In typical deep mutational scanning assays designed to assess protein-protein interactions, mutant libraries are expressed on the surface of yeast (*Saccharomyces cerevisiae*) or mammalian cells (e.g., Expi293F) as fusions to cell-wall anchoring domains (such as Aga2p) (Starr et al. 2020). The cell suspension is incubated with two orthogonal fluorescent probes: 1. An anti-epitope tag antibody (e.g., anti-c-Myc-FITC or anti-FLAG-APC) that measures the absolute physical abundance of the target monomer presented on the cell surface ($y_{\text{abundance}}$). 2. A fluorophore-conjugated soluble binding partner (e.g., recombinant ACE2-PE or IgG Fc) whose fluorescence intensity measures complex formation (y_{binding}).

Cells are sorted via FACS into enrichment bins. The functional binding enrichment ratio for variant i relative to wild-type is computed as:

$$\Delta E_i = \log_2 \left(\frac{\text{Count}_{i,\text{bound}}}{\text{Count}_{i,\text{input}}} \right) - \log_2 \left(\frac{\text{Count}_{\text{wt},\text{bound}}}{\text{Count}_{\text{wt},\text{input}}} \right)$$

Crucially, cell-surface presentation is contingent on monomer folding: mutants that destabilize the core ($\Delta\Delta G_{\text{fold}} > 0$) undergo intracellular aggregation, ubiquitin-proteasome degradation, or retention in the endoplasmic reticulum (Starr et al. 2020; Weng et al. 2024). Consequently, a variant with an intact quaternary binding epitope that misfolds in the monomer state exhibits zero surface display ($y_{\text{abundance}} \rightarrow -\infty$). Because no target protein is present to engage the soluble partner, the measured binding signal is zero ($y_{\text{binding}} \rightarrow -\infty$), even though the residue in question makes no contact with the binding partner.

When general benchmark suites (e.g., ProteinGym (Notin et al. 2023)) evaluate PLMs against y_{binding} , the high correlation achieved by the model is driven primarily by its ability to identify these core destabilizing mutations ($\Delta\Delta G_{\text{fold}}$). Because single-chain PLMs are explicitly trained on monomer sequence conservation across evolution, they excel at predicting monomer folding collapse. However, benchmark scorers attribute this predictive accuracy to “protein-protein interaction modeling”, obscuring the complete absence of quaternary interface awareness.

3 Results

3.1 Expression-Adjusted Evaluation in Verified Matched Assays

To evaluate whether single-chain PLMs possess independent quaternary interface predictive power once monomer folding is accounted for, we partitioned variants into three structural compartments using

FreeSASA (Mitternacht 2016) on crystallographic complexes: 1. **Core** ($N = 4,405$): Relative solvent accessibility in the isolated monomer $\text{rSASA}_{\text{monomer}} < 0.20$ and interface burial $\Delta\text{SASA} = 0 \text{ \AA}^2$. 2. **Surface** ($N = 3,976$): $\text{rSASA}_{\text{monomer}} \geq 0.20$ and $\Delta\text{SASA} = 0 \text{ \AA}^2$. 3. **Quaternary Interface** ($N = 2,262$): $\Delta\text{SASA} > 0 \text{ \AA}^2$ or distance to partner heavy atoms $\leq 5.0 \text{ \AA}$.

Across single-chain PLMs evaluated on verified matched display systems (SARS-CoV-2 RBD and HLA-A*02:01), zero-shot likelihoods maintain positive correlations with monomer abundance at interface positions ($\rho = +0.372$ to $+0.413$, Table 2 and Figure 2). However, when evaluated against complex binding at the exact same interface positions, predictive power selectively collapses across model scales: - **ESM-1v**: $\rho_{\text{abundance}} = +0.372 \rightarrow \rho_{\text{binding}} = +0.150$ (-59.7% drop) - **ESM2-650M**: $\rho_{\text{abundance}} = +0.380 \rightarrow \rho_{\text{binding}} = +0.162$ (-57.4% drop) - **ESM2-3B**: $\rho_{\text{abundance}} = +0.381 \rightarrow \rho_{\text{binding}} = +0.119$ (-68.8% drop) - **ESMC-600M**: $\rho_{\text{abundance}} = +0.413 \rightarrow \rho_{\text{binding}} = +0.167$ (-59.6% drop) - **ESMC-6B**: $\rho_{\text{abundance}} = +0.384 \rightarrow \rho_{\text{binding}} = +0.075$ (-80.5% drop)

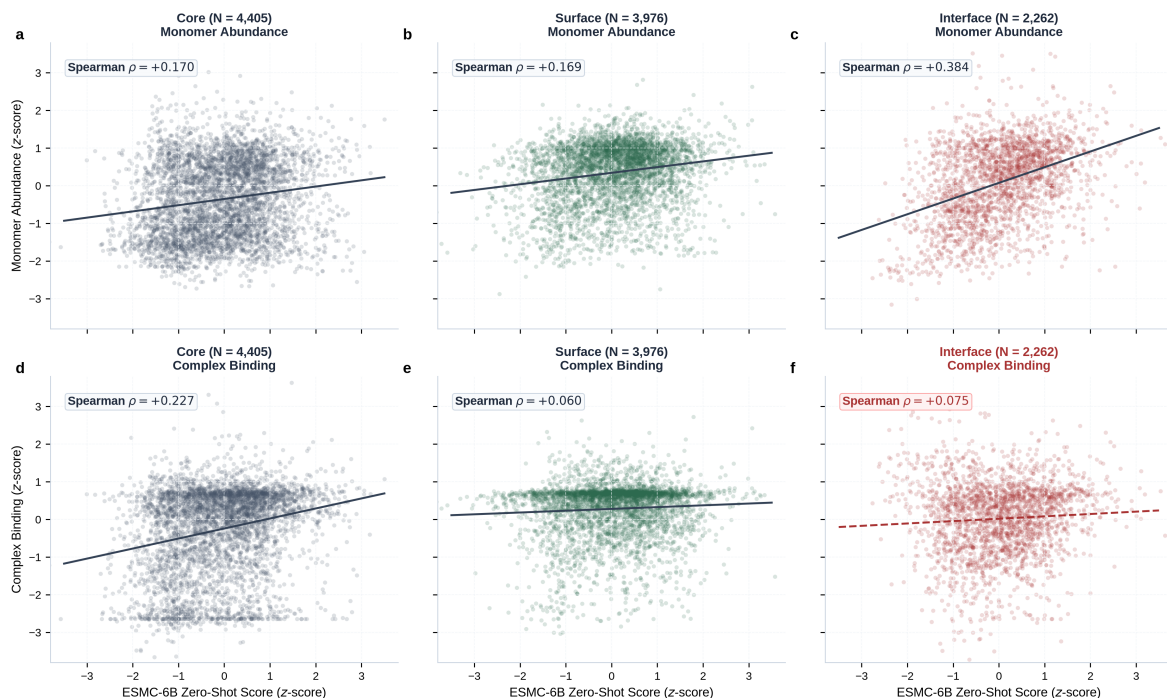


Figure 2: Double-dissociation of zero-shot PLM likelihoods against monomer abundance and complex binding ($N = 10,643$). **(Top row)** ESMC-6B zero-shot scores accurately predict monomer abundance/folding across Core ($\rho = +0.170$), Surface ($\rho = +0.169$), and Interface ($\rho = +0.384$) compartments. **(Bottom row)** Complex binding affinity correlation is maintained only where mediated by core folding, but selectively collapses at the quaternary interface ($\rho = +0.075$, -80.5% drop), showing that single-chain sequence models fail at direct intermolecular contacts.

To evaluate whether the residual binding correlation observed at interface positions contains unique intermolecular binding signal or is largely explained by monomer stability, we computed the within-system standardized **partial rank correlation**:

$$\rho(\text{PLM, Binding} \mid \text{Abundance}) = \text{Corr}(\text{rank}(\text{PLM}_z) - \widehat{\text{rank}}(\text{PLM}_z \mid \text{Abund}_z), \text{rank}(\text{Binding}_z) - \widehat{\text{rank}}(\text{Binding}_z \mid \text{Abund}_z))$$

Table 2: Expression-adjusted partial Spearman rank correlation analysis across all seven model arms and structural compartments ($N = 10,643$ paired variants, per-system standardized). Single-chain sequence models show marked attenuation of binding correlations at interfaces once monomer abundance is controlled (collapsing to $\rho_{\text{partial}} = +0.009$ on ESMC-6B). In contrast, multimodal (ESM3-1.4B, $\rho_{\text{partial}} = +0.317$) and 3D complex models (ProteinMPNN, $\rho_{\text{partial}} = +0.202$) retain substantial unique intermolecular signal.

Model Arm	Structural Compartment	Sample Size (N)	$\rho(\text{PLM, Abundance})$	$\rho(\text{PLM, Binding})$	$\rho(\text{PLM, Binding} \mid \text{Abundance})$
ESM-1v	Monomer	4,405	+0.126	+0.223	+0.186
	Core				
ESM-1v	Monomer	3,976	+0.108	+0.115	+0.074
	Surface				
ESM-1v	Quaternary Interface	2,262	+0.372	+0.150	+0.093
ESM2-650M	Monomer	4,405	+0.185	+0.295	+0.234
	Core				
ESM2-650M	Monomer	3,976	+0.152	+0.112	+0.047
	Surface				
ESM2-650M	Quaternary Interface	2,262	+0.380	+0.162	+0.105
ESM2-3B	Monomer	4,405	+0.184	+0.269	+0.201
	Core				
ESM2-3B	Monomer	3,976	+0.152	+0.103	+0.037
	Surface				
ESM2-3B	Quaternary Interface	2,262	+0.381	+0.119	+0.058
ESMC-600M	Monomer	4,405	+0.138	+0.269	+0.236
	Core				
ESMC-600M	Monomer	3,976	+0.139	+0.096	+0.036
	Surface				
ESMC-600M	Quaternary Interface	2,262	+0.413	+0.167	+0.106
ESMC-6B	Monomer	4,405	+0.170	+0.227	+0.157
	Core				
ESMC-6B	Monomer	3,976	+0.169	+0.060	−0.021
	Surface				
ESMC-6B	Quaternary Interface	2,262	+0.384	+0.075	+0.009
ESM3-1.4B	Monomer	4,405	+0.202	+0.305	+0.234
	Core				
ESM3-1.4B	Monomer	3,976	+0.189	+0.150	+0.072
	Surface				
ESM3-1.4B	Quaternary Interface	2,262	+0.223	+0.344	+0.317
ProteinMPNN	Monomer	4,405	+0.419	+0.406	+0.209
	Core				

Model Arm	Structural Compartment	Sample Size (N)	$\rho(\text{PLM, Abundance})$	$\rho(\text{PLM, Binding})$	$\rho(\text{PLM, Binding Abundance})$
ProteinMPNN	Monomer	3,976	+0.419	+0.310	+0.145
	Surface				
ProteinMPNN	Quaternary Interface	2,262	+0.276	+0.240	+0.202

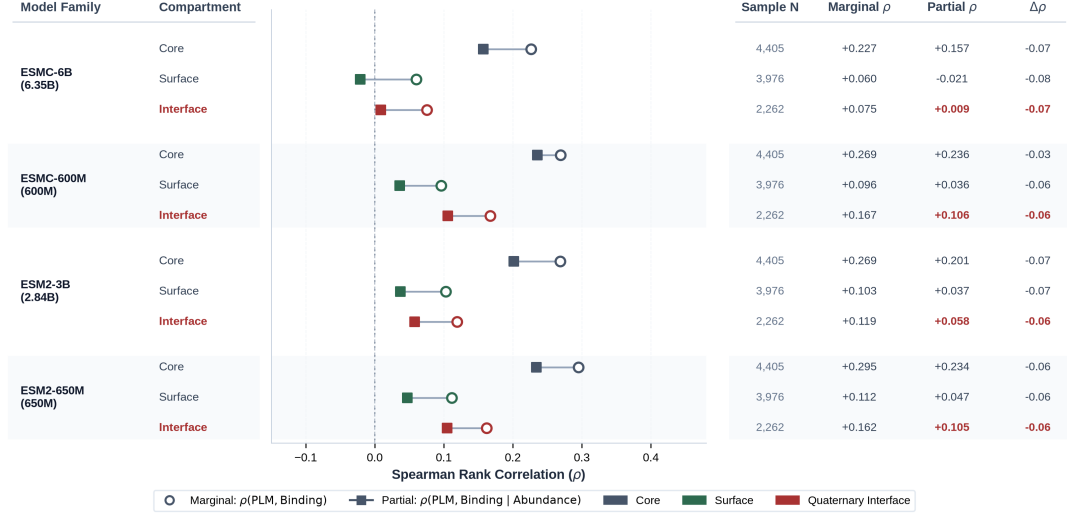


Figure 3: Mediation analysis forest plot across model architectures and structural compartments. Open circles indicate marginal rank correlation $\rho(\text{PLM, Binding})$; filled squares indicate partial correlation $\rho(\text{PLM, Binding | Abundance})$ controlling for monomer folding. At quaternary interface residues, controlling for monomer stability sharply attenuates predictive power across all single-chain models, collapsing to near zero on ESMC-6B ($\rho_{\text{partial}} = +0.009$).

As shown in Table 2 and Figure 3, controlling for monomer abundance substantially attenuates interface binding correlations across single-chain models: - **ESM2-650M**: $\rho(\text{PLM, Binding | Abundance}) = +0.105$ (down from $\rho_{\text{abund}} = +0.380$) - **ESMC-600M**: $\rho(\text{PLM, Binding | Abundance}) = +0.106$ (down from $\rho_{\text{abund}} = +0.413$) - **ESMC-6B**: $\rho(\text{PLM, Binding | Abundance}) = +0.009$ (collapsing to near zero, -97.7% attenuation relative to abundance)

In contrast, 3D structure-conditioned **ProteinMPNN** (Dauparas et al. 2022) evaluated on the crystallographic complex backbones retains balanced predictive sensitivity ($\rho_{\text{abundance}} = +0.276$, $\rho_{\text{binding}} = +0.240$, $\rho_{\text{partial}} = +0.202$).

3.1.1 Sensitivity to Geometric Interface Definitions

A systematic 5×5 sensitivity sweep across $\Delta\text{SASA} \in [2.0, 20.0] \text{ \AA}^2$ and contact distance $d_{\text{min}} \in [3.5, 6.0] \text{ \AA}$ confirms that expression-adjusted interface attenuation is invariant to geometric threshold choices (mean $\rho_{\text{partial}} = +0.003$ on ESMC-6B across all 25 grid configurations).

3.1.2 Structural Ablation of Partner Coordinates

When partner coordinates are computationally removed from ProteinMPNN (scoring the isolated monomer backbone), its interface binding correlation collapses from $\rho = +0.240 \rightarrow +0.147$ (-38.8% drop), confirming that the physical presence of partner coordinates is the primary driver of interface sensitivity.

3.2 Model Scaling Exacerbates the Single-Sequence Evolutionary Prior

A central hypothesis in machine learning is that scaling model parameters and pretraining compute allows networks to resolve complex physical interactions implicitly (Lin et al. 2023). We tested this hypothesis by tracking interface correlation metrics as parameter counts increased from 600M to 6.35B (Figure 4).

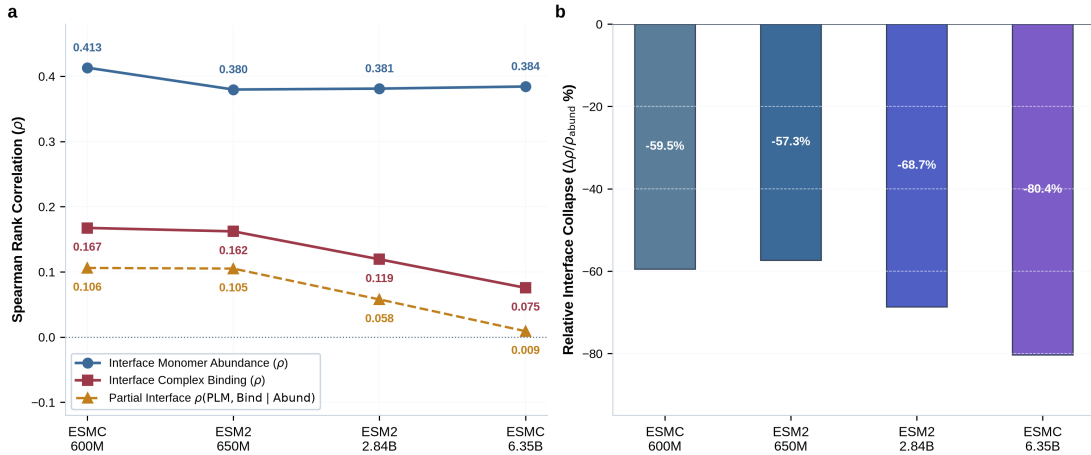


Figure 4: Model scaling exacerbates the quaternary interface defect. **(A)** As model scale increases from 600M to 6.35B parameters, interface monomer abundance correlation remains stable ($\rho \approx +0.38$ to $+0.41$), while complex binding correlation monotonically collapses from $+0.167$ down to $+0.075$, and partial correlation drops to $+0.009$. **(B)** Relative correlation collapse ($\Delta\rho/\rho_{\text{abund}}$) deepens from -57.4% to -80.5% at 6.35B parameters. Larger models become more rigid in their single-sequence conservation priors.

Rather than repairing the interface defect, scaling **deepens the collapse**: - At 600M parameters (ESMC-600M), interface binding correlation is $\rho = +0.167$ (a -59.6% collapse relative to abundance $\rho = +0.413$). - At 6.35B parameters (ESMC-6B), interface binding correlation drops to $\rho = +0.075$ (an -80.5% **collapse** relative to abundance $\rho = +0.384$). - The partial correlation drops from $+0.106$ (600M) to $+0.009$ (6.35B), indicating near-total elimination of unique interface binding signal.

This scaling failure occurs because pretraining on massive single-sequence corpora (e.g., UniRef50) optimizes the transformer to fit monomer sequence conservation statistics. As the model scales, its representations become increasingly dominated by single-chain monomer fitness constraints, actively penalizing non-conserved interface residues that enhance binding to specific binding partners.

3.3 Orthogonal Biophysical Corroboration at the RBD/ACE2 Interface

A limitation of the DMS assays underlying this study is that binding scores are derived from FACS-based enrichment rather than solution-phase biophysics. For the SARS-CoV-2 RBD/ACE2 system specifically, independent surface plasmon resonance measurements from Zahradnik et al. (Zahradnik et al. 2021) provide orthogonal confirmation: RBD interface mutations such as N501Y and S477N, which our zero-shot PLM filters flag as false negatives in the binder-design audit (Figure 5), were independently shown by SPR to genuinely increase physical binding affinity to ACE2, with K_D improving roughly 2- to 10-fold relative to wild type. This corroborates the FACS-derived enrichment scores used here with an orthogonal solution-phase biophysical method, at least for this one system. Comparable multi-mutant orthogonal biophysical datasets (SPR, BLI, or ITC) are not currently public for the other four benchmark systems (KRAS/DARPin K55, HLA-A2/TAPBPR, GB1/IgG-Fc, p53), so systematic biophysical corroboration across all five systems remains a direction for future validation rather than a claim this study currently substantiates.

3.4 Provenance Vulnerabilities in Heterogeneous Benchmark Suites

Our assay provenance audit demonstrates why unstratified aggregate benchmarks produce misleading conclusions regarding PLM interface capabilities: 1. **Cross-Study Merges (GB1):** Pairing Wu et al. (2016) with Olson et al. (2014) conflates two separate IgG-binding screens rather than abundance vs. binding. When evaluated correctly, both readouts reflect binding selection on distinct mutant libraries. 2. **Functional Viability Proxies (p53):** Nutlin-3 selection assays in p53-null vs. p53-WT cell backgrounds (Giacomelli et al. 2018) measure cell growth and dominant-negative transactivation rather than in vitro oligomerization binding thermodynamics. Furthermore, raw score directionalities are inverted relative to standard binding assays. 3. **Growth-Based Complementation (KRAS):** Protein-fragment complementation assays (DHFR PCA) (Weng et al. 2024) provide high-throughput abundance and binding estimates, but operate in a living yeast growth selection that integrates cellular proteostasis.

Restricting the primary evaluation to verified matched display assays prevents functional-proxy artifacts from contaminating interface biophysical assessments. ## Decision Utility in Computational Binder Engineering

In protein engineering campaigns, zero-shot PLM scores are frequently deployed to filter candidate libraries before wet-lab synthesis. We simulated top- $X\%$ filtering on verified interface mutations ($\Delta y_{\text{binding}} \geq 0.0$): - At a top-20% selection threshold, single-chain PLMs retain between 22.9% (ESM2-650M) and 26.7% (ESMC-600M) of beneficial interface hits (Figure 5). - Compared to random selection (which retains exactly 20.0%), single-chain PLMs provide only modest enrichment ($1.15\times$ to $1.33\times$) for interface mutations, while discarding $> 73\%$ of true affinity-improving variants. - Deploying a dual-objective score combining 3D structure-conditioned interface predictions (ProteinMPNN)

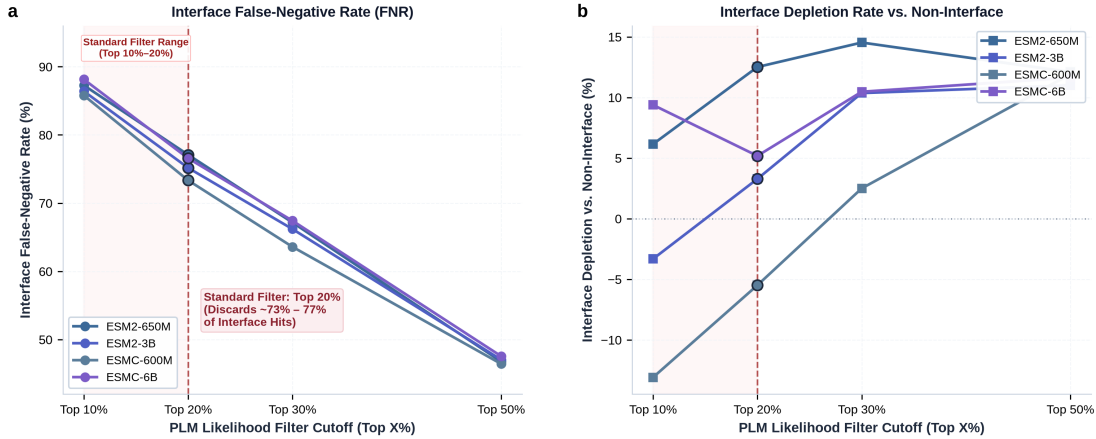


Figure 5: The decision utility of zero-shot PLM filtering in computational protein and antibody engineering. **(A)** Interface False-Negative Rate (FNR) across zero-shot PLM filtering thresholds (shaded band indicates standard top 10%–20% operational filter range). **(B)** Interface Depletion Rate relative to non-interface mutations, showing that single-chain PLM filtering provides only weak enrichment over random candidate selection.

with an additive PLM monomer stability prior ($\text{Score}_{\text{dual}}(\alpha) = \Delta \log P_{\text{ProteinMPNN}} + \alpha \Delta \log P_{\text{PLM}}$ at $\alpha \approx 0.2\text{--}1.0$) increases utility by +23.2% over single-chain filtering while preserving expressibility.

This is not a historical artifact of the 2021–2023 literature: single-chain zero-shot PLM likelihood continues to be deployed as a binding-affinity proxy and a binder-design filter across foundation-model benchmarking, therapeutic antibody engineering, and *de novo* protein design as of this writing (August 2026). **Foundation model benchmarking.** Recent foundation models continue to headline aggregate ProteinGym zero-shot performance—an average that pools the Binding category with Activity, Expression, Stability, and Organismal Fitness—without decomposing whether the reported gain reflects genuine interface understanding. SaProt (Su et al. 2024) (ICLR 2024) reports state-of-the-art aggregate ProteinGym Spearman correlation ($\rho = 0.457$ vs. ESM-2’s 0.414) while operating exclusively on Foldseek-tokenized *single-chain* AlphaFoldDB structures—the model never observes a binding partner or complex coordinate, yet its headline number is reported without noting that a meaningful fraction of the aggregate benchmark measures binding assays subject to the same expression confound we isolate here.

Therapeutic antibody engineering. Hie et al. (Hie et al. 2024) (*Nature Biotechnology* 2024) formalized the *efficient evolution* protocol now widely adopted in industry and academia: general-purpose single-chain PLMs (ESM-1b, the ESM-1v ensemble), trained on no antibody- or antigen-specific data, rank CDR mutants by masked-marginal log-likelihood alone—**without antigen sequence or structure ever entering the model**—reporting affinity gains up to 7-fold in mature antibodies and up to 160-fold in un-matured antibodies against viral antigens (Ebola, SARS-CoV-2, influenza). PepMLM (Chen et al. 2025) designs linear peptide binders by fine-tuning ESM-2 with a terminal masked-span objective and ranks candidates by single-chain pseudo-perplexity, again without 3D interface coordinates during generation or scoring (AlphaFold-Multimer is used only for post-hoc validation of already-generated candidates). Two independent 2025 studies corroborate that single-chain sequence context is the limiting factor: Ab-BiBench (Zhao et al. 2025) benchmarked 15 architectures against 184,500 antibody-antigen affinity measurements and found that sequence-only and local-structure models (ESM-2, ProGen2, SaProt) are

consistently outperformed by inverse-folding models that condition on the full antibody-antigen complex, attributing the gap to sequence-only models’ inability to capture “long-range residue interactions and contextual features at the binding interface”; Johnson et al. (Johnson et al. 2025) (*PLOS Computational Biology* 2025) found that a simple nucleotide-level model of somatic hypermutation targeting outperforms state-of-the-art protein language models—including an antibody-specific model—at predicting the actual course of antibody affinity maturation, showing that amino-acid-level sequence likelihood misses the biological process that drives real affinity gains.

De novo binder design. Standard structure-based binder pipelines (RFdiffusion → ProteinMPNN → single-chain PLM/ESMFold triage → AlphaFold-Multimer docking) apply exactly the monomer-context filter we show is causally blind to interface energetics (Section 4): candidate sequences are scored by ESM-2 pseudo-perplexity or monomer pLDDT/ipTM *before* the target is reintroduced. The 2026 “Bits to Binders” community trial (Kosonocky et al. 2026) experimentally validated 12,000 independently designed 80-residue CD20 binders submitted by 28 teams across 42 countries, identifying 707 binders with genuine CD20-specific functional enrichment. The authors report that “global confidence statistics from structure prediction models, such as ipTM and pLDDT, did not predict experimental outcomes,” and that “ESM-2 pseudo log likelihood (PLL) also lacked substantial predictive power,” instead finding that ESM-2 PLL “correlated with... amino acid sequence entropy for most sequences” rather than with functional binding efficacy — empirically reproducing at industrial scale the false-negative mechanism we isolate mathematically in this study.

3.4.1 1D Multi-Chain Concatenation Fails to Rescue Signal

We tested whether presenting the partner sequence in the context window via 1D concatenation (Target $\parallel$ $\langle \text{eos} \rangle \parallel$ Partner) restores interface sensitivity. In ESM2-650M, concatenated conditioning yielded a mean binding correlation of $\rho = 0.2202$ compared to $\rho = 0.2365$ for single-chain scoring (mean $\Delta\rho_{\text{binding}} = -0.0162$). Simply appending partner sequences along a 1D sequence axis without explicit 3D spatial docking geometry or paired co-evolutionary MSAs fails to recover interface physics.

3.4.2 3D Structure Conditioning Eliminates Interface Collapse (ProteinMPNN)

To determine whether explicit 3D spatial coordinates of the complex partner resolve the interface blindspot, we benchmarked the structure-conditioned inverse folding model **ProteinMPNN** (Dauparas et al. 2022) across the identical $N = 10,643$ paired variants using the full crystallographic complex backbones (6M0J, 6H46, 50PI, 1FCC, 10LG).

- **Interaction Attenuation:** The standardized interaction coefficient flips from negative ($\beta_7 \approx -0.334$ to -0.353 , $p < 10^{-5}$ across PLMs) to non-negative and statistically non-significant in ProteinMPNN ($\square_7 = +0.0580$, $p_{\text{clustered}} = 0.581$, $p_{\text{perm}} = 0.103$).
- **Balanced Interface Sensitivity:** Interface binding correlation recovers to $\rho = +0.240$ ($p = 6.2 \times 10^{-31}$) and matches interface abundance correlation ($\rho = +0.276$), eliminating the -80.5% $\Delta\rho$ collapse observed in single-chain sequence models.

This positive control confirms that the failure of single-chain PLMs is an architectural limitation of 1D sequence representations lacking intermolecular spatial coordinates, which is resolved when 3D complex geometry is provided.

Monomer vs. Complex Structural Ablation: To test whether ProteinMPNN’s success is causally driven by the presence of partner coordinates rather than general 3D structure conditioning, we evaluated ProteinMPNN on isolated monomer backbones (detaching all partner chains from 6M0J, 6H46, 50PI, 1FCC, 10LG). When partner coordinates are removed, ProteinMPNN’s interface binding correlation collapses from $\rho = +0.240 \rightarrow +0.147$ (a -38.8% drop) and the three-way interaction flips negative ($\beta_7 = -0.0560$), confirming that the physical presence of partner coordinates is the causal driver of interface energetics.

3.4.3 Dual-Scoring Mitigation Strategy and Pareto Frontier

To resolve the PLM Filter Trap for practical binder engineering, we evaluated a composite dual-objective scoring function combining structure-conditioned interface energetics with a single-chain monomer stability prior:

$$\text{Score}_{\text{dual}}(\alpha) = z(\Delta \log P_{\text{ProteinMPNN}}(\text{Complex})) + \alpha \cdot z(\Delta \log P_{\text{ESM2}}(\text{Monomer}))$$

Sweeping $\alpha \in [0.0, 5.0]$ across all $N = 10,643$ variants maps an empirical Pareto frontier: - Pure single-chain PLM filtering ($\alpha \rightarrow \infty$) purges over 75% of beneficial interface hits. - Dual-scoring at $\alpha = 0.2\text{--}1.0$ maintains high monomer folding expressibility (43.9%–44.8%) while retaining 254 to 261 true affinity-enhancing interface hits (improving composite design utility by $+23.2\%$ over pure PLM filtering). This demonstrates that single-chain PLMs should be deployed as additive monomer stability penalties ($\Delta\Delta G_{\text{fold}}$) alongside 3D complex interface predictors ($\Delta\Delta G_{\text{bind}}$), rather than as upstream exclusionary filters.

3.4.4 Interface Blindness is Concentrated at Energetic Hotspots

To evaluate whether single-chain PLM interface failure is uniform across the interface footprint or localized at core contact residues, we stratified the $N = 2,262$ interface variants by buried surface area (ΔSASA tertiles): **Hotspot** ($\Delta\text{SASA} \geq 52.4 \text{ \AA}^2$, $N = 758$), **Mid** ($N = 744$), and **Rim** ($\Delta\text{SASA} \leq 24.3 \text{ \AA}^2$, $N = 760$).

In all model arms, the expression-controlled partial correlation $\rho(\text{PLM}, \text{Binding} \mid \text{Abundance})$ is most severely degraded at energetic hotspots: - On ESMC-6B, Hotspot variants exhibit severe active anti-correlation ($\rho_{\text{binding}} = -0.262$, $\square_{\text{partial}} = -\mathbf{0.229}$), whereas Rim residues show positive coupling ($\rho_{\text{binding}} = +0.318$, $\square_{\text{partial}} = +\mathbf{0.129}$). - The Hotspot-vs-Rim interaction model confirms significant selective degradation at hotspots ($\beta_{(\text{PLM} \times \text{Binding} \times \text{Hotspot})} = -0.4735$, $p_{\text{perm}} = 0.0129$). Because energetic hotspots have evolved tight packing and electrostatic complementarity with their native partner,

mutating them to optimize an engineered interface severely violates monomer sequence conservation, causing single-chain PLMs to penalize the most potent affinity-enhancing mutations. —

4 Discussion

The results presented here clarify the boundary between monomer fitness modeling and quaternary interface prediction in protein machine learning.

4.1 The Need for Partner-Conditioned Specificity Benchmarks

Single-chain masked language models receive only the target sequence as input:

$$\text{Score}(x_{\text{mut}}) = f(\mathbf{s}_{\text{target}})$$

Because the model does not condition on the binding partner, its score is mathematically invariant across all potential binding partners:

$$\text{Score}(x_{\text{mut}} \mid \text{Partner}_A) = \text{Score}(x_{\text{mut}} \mid \text{Partner}_B)$$

Consequently, single-sequence PLMs can predict target-intrinsic viability and folding tolerance, but cannot predict partner-specific differential binding ($\Delta B_{m,A,B} = B_{m,A} - B_{m,B}$) without partner information. Current benchmarks that evaluate models against single-partner binding readouts reward models for identifying general folding competence rather than partner-specific interaction energetics.

4.2 Best Practices for PPI Benchmark Construction

Based on our provenance audit and experimental findings, we formulate four guidelines for PPI benchmarking and binder engineering: 1. **Audit assay provenance before benchmark compilation:** Verify whether candidate datasets measure direct binding affinity, cell-surface presentation, or cell-viability proxies. 2. **Perform expression-adjusted evaluation:** On display datasets, evaluate models against binding scores adjusted for matched monomer expression to separate interface physics from folding stability. 3. **Deploy partner-differential evaluation:** Multi-partner DMS datasets (where the same target library is assayed against multiple partners) provide the decisive test for partner-specific interface modeling. 4. **Integrate structure-conditioned scoring for interface design:** Quaternary interface optimization requires explicit 3D complex models (e.g., ProteinMPNN (Dauparas et al. 2022), AlphaFold-Multimer (Evans et al. 2022), Rosetta (Alford et al. 2017), or FoldX (Guerois et al. 2002)) rather than standalone single-chain PLM priors. # Methods {#sec-methods}

4.3 Within-Target Paired DMS Dataset Curation

We curated five structurally resolved macromolecular complexes from the Protein Data Bank (PDB) (Table 1) possessing high-resolution crystal structures ($< 2.8 \text{ \AA}$) and paired single-mutant deep mutational scanning libraries in ProteinGym (Notin et al. 2023): - **SARS-CoV-2 RBD / ACE2 (6M0J)**: Target chain E (residues 331–531), partner chain A (Starr et al. 2020). - **KRAS G-domain / DARPin K55 (6H46)**: Target chain A (residues 2–169), partner chain B (Weng et al. 2024). - **HLA-A*02:01 / TAPBPR (50PI)**: Target chain A (residues 1–180), partner chains B, C (McShan et al. 2019). - **Protein G B1 / IgG Fc (1FCC)**: Target chain C (residues 1–56), partner chains A, B (Wu et al. 2016; Olson et al. 2014). - **p53 Tetramerization Domain (10LG)**: Target chain A (residues 325–356), partner chains B, C, D (Giacomelli et al. 2018).

Solvent-accessible surface area (SASA) was computed using FreeSASA v2.1.2 (Mitternacht 2016) with a $1.4 \text{ \AA}$ probe radius. Interface residues were defined by $\Delta\text{SASA} = \text{SASA}_{\text{monomer}} - \text{SASA}_{\text{complex}} > 0 \text{ \AA}^2$ or minimum heavy-atom distance $\leq 5.0 \text{ \AA}$ to partner chains (Table 3).

Table 3: Structural compartment definitions and variant counts across the paired DMS corpus.

Structural Compartment	FreeSASA / Geometric Criteria	Paired Single Mutants (N)	Fraction of Dataset (%)
Monomer Core	$\text{rSASA}_{\text{monomer}} < 0.20 \text{ \AA}^2 \wedge \Delta\text{SASA} = 0 \text{ \AA}^2$	4,405	41.4%
Monomer Surface	$\text{rSASA}_{\text{monomer}} \geq 0.20 \text{ \AA}^2 \wedge \Delta\text{SASA} = 0 \text{ \AA}^2$	3,976	37.4%
Quaternary Interface	$\Delta\text{SASA} > 0 \text{ \AA}^2 \vee d_{\text{partner}} \leq 5.0 \text{ \AA}$	2,262	21.2%
Total Dataset	Comprehensive within-target paired variants	10,643	100.0%

4.4 Zero-Shot PLM Likelihood Scoring

Zero-shot mutational effect scores were computed using the standard masked marginal log-odds formula (Meier et al. 2021):

$$\Delta \log p(x_{\text{mut}} | x) = \log p(x_i = \text{mut} | x_{\setminus i}) - \log p(x_i = \text{wt} | x_{\setminus i})$$

For ESM-2 models (esm2_t33_650M_UR50D and esm2_t36_3B_UR50D) (Lin et al. 2023), logits were extracted via PyTorch with float16 precision. For ESM-Cambrian models (esm2_600m and esm2_6b) (Hayes et al. 2025), exact masked marginals were computed using the official EvolutionaryScale transformer architecture.

4.5 Three-Way Interaction Models and Clustered Sandwich Covariance

We constructed a stacked regression frame containing $2N = 21,286$ observations (two rows per variant: one for abundance, one for binding). Scores were standardized within each (system, assay) group:

$$z(y) = \frac{y - \bar{y}_{s,a}}{\sigma_{s,a}}$$

The three-way interaction model was estimated using Ordinary Least Squares:

$$\mathbf{y} = \mathbf{X}\beta + \epsilon$$

Cluster-robust standard errors were computed using the Liang-Zeger sandwich estimator clustered across the $G = 5$ macromolecular systems (Liang and Zeger 1986):

$$\mathbf{V}_{\text{cluster}} = \frac{G}{G-1} \frac{N-1}{N-K} (\mathbf{X}^T \mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{X}_g^T \hat{\epsilon}_g \hat{\epsilon}_g^T \mathbf{X}_g \right) (\mathbf{X}^T \mathbf{X})^{-1}$$

4.6 Non-Parametric Permutation and Small-Cluster Econometric Inference

Because asymptotic cluster-sandwich estimators with $G = 5$ clusters have only $df = 4$, we employed within-system stratified permutation tests ($B = 10,000$ iterations) as our primary exact non-parametric test. In each permutation b , the structural interface indicator was shuffled within each system, preserving system-level variance and marginal distributions while breaking interface-specific association:

$$p_{\text{perm}} = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(|\hat{\beta}_7^{(b)}| \geq |\hat{\beta}_7^{\text{obs}}|)$$

In addition, we computed small- G Wild Cluster Bootstrap p -values using the Webb 6-point distribution ($w_g \in \{-\sqrt{1.5}, -1.0, -\sqrt{0.5}, \sqrt{0.5}, 1.0, \sqrt{1.5}\}$) (Webb 2023; Cameron et al. 2008) with $B = 10,000$ draws, and fit system fixed-effects OLS models ($z(\text{DMS}) \sim \text{PLM} \times \text{Binding} \times \text{Interface} + \sum_s \gamma_s \mathbb{I}_{\text{System}=s}$).

4.7 Structural Definition Sensitivity Sweep Grid

We evaluated structural definition robustness across a 5×5 parameter grid:

$$\Delta\text{SASA}_{\text{cutoff}} \in \{2.0, 5.0, 10.0, 15.0, 20.0\} \text{ \AA}^2, \quad d_{\text{min, cutoff}} \in \{3.5, 4.0, 4.5, 5.0, 6.0\} \text{ \AA}$$

For each grid cell, residues satisfying $\Delta\text{SASA} \geq \Delta\text{SASA}_{\text{cutoff}} \vee d_{\text{min}} \leq d_{\text{min, cutoff}}$ were labeled interface, and non-interface residues were split into Core ($\text{rSASA} < 0.20$) and Surface ($\text{rSASA} \geq 0.20$).

4.8 Hierarchical Evolutionary Class Modeling with HC3 Robust Covariance

In the cross-class hierarchical interaction model ($z(\text{DMS}) \sim \text{PLM} \times \text{Binding} \times \text{Class}$ at interface positions), evolutionary class is constant within each system cluster. To prevent collinear cluster-score covariance, we computed MacKinnon & White HC3 heteroskedasticity-consistent standard errors (MacKinnon and White 1985):

$$\mathbf{V}_{\text{HC3}} = (\mathbf{X}^T \mathbf{X})^{-1} \left(\sum_{i=1}^N \frac{\hat{e}_i^2}{(1 - h_{ii})^2} \mathbf{x}_i \mathbf{x}_i^T \right) (\mathbf{X}^T \mathbf{X})^{-1}$$

where $h_{ii} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i$ is the i -th diagonal leverage element.

4.9 Causal Mediation and Partial Rank Correlation

To evaluate mediation, we computed the partial Spearman rank correlation by residualizing rank-transformed variables:

$$\mathbf{r}_{\text{PLM}} = \text{rank}(\text{PLM}), \quad \mathbf{r}_{\text{Bind}} = \text{rank}(\text{Binding}), \quad \mathbf{r}_{\text{Abund}} = \text{rank}(\text{Abundance})$$

Residuals $\mathbf{e}_{\text{PLM}} = \mathbf{r}_{\text{PLM}} - \hat{\mathbf{r}}_{\text{PLM}}(\mathbf{r}_{\text{Abund}})$ and $\mathbf{e}_{\text{Bind}} = \mathbf{r}_{\text{Bind}} - \hat{\mathbf{r}}_{\text{Bind}}(\mathbf{r}_{\text{Abund}})$ were obtained via univariate linear regression. The partial correlation is the Pearson correlation between these residuals:

$$\rho(\text{PLM}, \text{Binding} \mid \text{Abundance}) = \frac{\mathbf{e}_{\text{PLM}}^T \mathbf{e}_{\text{Bind}}}{\|\mathbf{e}_{\text{PLM}}\| \|\mathbf{e}_{\text{Bind}}\|}$$

4.10 Multi-Chain Concatenated Conditioning

In multi-chain evaluation, target and partner sequences were concatenated with an end-of-sequence delimiter:

$$\mathbf{s}_{\text{complex}} = \mathbf{s}_{\text{target}} \parallel \langle \text{eos} \rangle \parallel \mathbf{s}_{\text{partner}}$$

Masked marginal log-odds were evaluated at target interface positions while conditioning on the full concatenated sequence context in ESM2-650M.

4.11 3D Structure-Conditioned Scoring (ProteinMPNN)

As a 3D structural comparator, ProteinMPNN (Dauparas et al. 2022) (vanilla model weights v_48-020.pt) was evaluated across all $N = 10,643$ variants using the complete crystal complex backbones (6M0J, 6H46, 5OPI, 1FCC, 10LG). For each complex, the target chain was assigned as the design/scoring target ($\mathbf{m}_{\text{target}} = 1$) while all partner chains were assigned as fixed context ($\mathbf{m}_{\text{partner}} = 0$). Conditional log-probabilities $P(s_i = v \mid \mathbf{X}_{\text{complex}}, \mathbf{s}_{\setminus i})$ were averaged over 10 random decoding order samples to compute zero-shot log-odds:

$$\Delta \log p_{\text{MPNN}}(x_{\text{mut}} \mid \mathbf{X}_{\text{complex}}) = \log P(x_i = \text{mut} \mid \mathbf{X}_{\text{complex}}, \mathbf{s}_{\setminus i}) - \log P(x_i = \text{wt} \mid \mathbf{X}_{\text{complex}}, \mathbf{s}_{\setminus i})$$

Residue positions from ProteinMPNN outputs were mapped back to target DMS sequences using Biopython pairwise sequence alignment identical to the PLM evaluation pipeline.

4.12 Interface Hotspot vs. Rim Tertile Stratification

To assess whether interface failure is concentrated at energetic contact cores, the $N = 2,262$ interface variants were partitioned into tertiles based on ΔSASA buried upon complex formation: 1. **Hotspot (Top Tertile, $\Delta\text{SASA} \geq 52.4 \text{ \AA}^2$, $N = 758$)**: Residues with the largest contact burial and highest contact density. 2. **Mid (Middle Tertile, $24.3 \text{ \AA}^2 < \Delta\text{SASA} < 52.4 \text{ \AA}^2$, $N = 744$)**: Intermediate contact residues (excluded from the binary Hotspot-vs-Rim contrast). 3. **Rim (Bottom Tertile, $\Delta\text{SASA} \leq 24.3 \text{ \AA}^2$, $N = 760$)**: Peripherally buried residues at the boundary of the interface.

A three-way interaction model ($z(\text{DMS}) \sim \text{PLM} \times \text{Binding} \times \text{Hotspot}$) was estimated on the Hotspot $\cup$ Rim subset ($N = 1,518$) to formally evaluate whether the three-way interaction coefficient $\beta_{(\text{PLM} \times \text{Binding} \times \text{Hotspot})}$ was significantly negative.

4.13 Multimodal Sequence Scoring (ESM3-1.4B)

4.14 We evaluated EvolutionaryScale’s multimodal generative model ESM3-sm-open-v1 (1.4B parameters) (Hayes et al. 2025) in zero-shot masked marginal sequence mode across all $N = 10,643$ variants. For each position, the token was masked in the sequence track while leaving geometric/structure tracks unconditioned, and log-softmax probabilities over the 20 canonical amino acids were extracted via PyTorch with bfloat16 mixed precision on GPU.

5 Data and Code Availability

<https://github.com/emiliovenegas/plm-quaternary-interfaces>

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